PURITY, FINITENESS OF THE SUPPORT AND UNIVERSAL PURITY FOR A SEPARATED FINITE TYPE MODULE WITH FINITE FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The scheme theoretic support of a finite type module and quasi-finiteness. Morphisms, labels morphisms-definition-scheme-theoretic-support (proof line 5063) and morphisms-lemma-quasi-finite (proof line 5066); Same chapter flat.tex, label lemma-supported-on-closed (proof line 5068). Reduces the statement to the scheme theoretic support Z, which is separated and quasi-finite over S, so that one may assume the support of the module is all of X.
- Purity of a proper or finite morphism and the relation between finiteness and properness. Same chapter flat.tex, label lemma-proper-pure (proof line 5076); Morphisms, label morphisms-lemma-finite-proper (proof line 5078). Gives that (2) implies (3) directly, since a finite morphism is proper and a proper morphism is pure relative to its base.
- Factoring a quasi-finite separated morphism through a finite one, and lifting specializations along an etale morphism. More on Morphisms, labels more-morphisms-lemma-quasi-finite-separated-pass-through-finite (proof line 5082) and more-morphisms-lemma-etale-splits-off-quasi-finite-part-technical (proof line 5100); Schemes, label schemes-lemma-quasi-compact-closed (proof line 5095); Morphisms, labels morphisms-lemma-generalizations-lift-flat and morphisms-lemma-etale-flat (proof lines 5110-5111). Supplies the factorisation j : X -> T followed by a finite pi, the criterion that a quasi-compact open immersion is closed as soon as specializations lift along it, the etale splitting of the quasi-finite part over an elementary etale neighbourhood of the base, and the lifting of specializations along the resulting etale morphism.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 05K3: sources/stacks-flat/pinned-original-source.tex, source lines 5016–5016. Navigation pointer for \ref{lemma-proper-pure} used inside the counted span (line 5076); the argument that consumes it is retained in the span.

Source reference. Original lemma 05K1: sources/stacks-flat/pinned-original-source.tex, source lines 4989–4989. Navigation pointer for \ref{lemma-supported-on-closed} used inside the counted span (line 5068); the argument that consumes it is retained in the span.

Source reference. Original lemma 02LN: sources/stacks-chow/tags.txt, source lines 3414–3414. Navigation pointer for \ref{more-morphisms-lemma-etale-splits-off-quasi-finite-part-technical} used inside the counted span (line 5100); the argument that consumes it is retained in the span.

Source reference. Original lemma 05K0: sources/stacks-chow/tags.txt, source lines 7184–7184. Navigation pointer for \ref{more-morphisms-lemma-quasi-finite-separated-pass-through-finite} used inside the counted span (line 5082); the argument that consumes it is retained in the span.

Source reference. Original lemma 05JV: sources/stacks-chow/tags.txt, source lines 7179–7179. Navigation pointer for \ref{morphisms-definition-scheme-theoretic-support} used inside the counted span (line 5063); the argument that consumes it is retained in the span.

Source reference. Original lemma 02GS: sources/stacks-chow/tags.txt, source lines 3239–3239. Navigation pointer for \ref{morphisms-lemma-etale-flat} used inside the counted span (line 5111); the argument that consumes it is retained in the span.

Source reference. Original lemma 01WN: sources/stacks-chow/tags.txt, source lines 2488–2488. Navigation pointer for \ref{morphisms-lemma-finite-proper} used inside the counted span (line 5078); the argument that consumes it is retained in the span.

Source reference. Original lemma 03HV: sources/stacks-chow/tags.txt, source lines 4581–4581. Navigation pointer for \ref{morphisms-lemma-generalizations-lift-flat} used inside the counted span (line 5110); the argument that consumes it is retained in the span.

Source reference. Original lemma 02NH: sources/stacks-chow/tags.txt, source lines 3480–3480. Navigation pointer for \ref{morphisms-lemma-quasi-finite} used inside the counted span (line 5066); the argument that consumes it is retained in the span.

Source reference. Original lemma 01K9: sources/stacks-chow/tags.txt, source lines 2042–2042. Navigation pointer for \ref{schemes-lemma-quasi-compact-closed} used inside the counted span (line 5095); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a separated, finite type morphism of schemes. Let $\mathcal{F}$ be a finite type, quasi-coherent $\mathcal{O}_X$-module. Assume that $Supp(\mathcal{F}_s)$ is finite for every $s \in S$. Then the following are equivalent*

(1) *$\mathcal{F}$ is pure relative to S,*
(2) *the scheme theoretic support of $\mathcal{F}$ is finite over S, and*
(3) *$\mathcal{F}$ is universally pure relative to S.*

In particular, given a quasi-finite separated morphism $X \to S$ we see that X is pure relative to S if and only if $X \to S$ is finite.

Proof. Let $Z \subset X$ be the scheme theoretic support of $\mathcal{F}$, see Morphisms, Definition 05JV. Then $Z \to S$ is a separated, finite type morphism of schemes with finite fibres. Hence it is separated and quasi-finite, see Morphisms, Lemma 02NH. By Lemma 05K1 it suffices to prove the lemma for $Z \to S$ and the sheaf $\mathcal{F}$ viewed as a finite type quasi-coherent module on Z. Hence we may assume that $X \to S$ is separated and quasi-finite and that $\mathrm{Supp}(\mathcal{F}) = X$.

It follows from Lemma 05K3 and Morphisms, Lemma 01WN that (2) implies (3). Trivially (3) implies (1). Assume (1) holds. We will prove that (2) holds. It is clear that we may assume S is affine. By More on Morphisms, Lemma 05K0 we can find a diagram

$$\begin{array}{ccc} X & \xrightarrow{\quad j \quad} & T \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle \pi} & \\ & S & \end{array}$$

with π finite and j a quasi-compact open immersion. If we show that j is closed, then j is a closed immersion and we conclude that $f = \pi \circ j$ is finite. To show that j is closed it suffices to show that specializations lift along j, see Schemes, Lemma 01K9. Let $x \in X$, set $t' = j(x)$ and let $t' \rightsquigarrow t$ be a specialization. We have to show $t \in j(X)$. Set $s' = f(x)$ and $s = \pi(t)$ so $s' \rightsquigarrow s$. By More on Morphisms, Lemma 02LN we can find an elementary étale neighbourhood $(U, u) \to (S, s)$ and a decomposition

$$T_U = T \times_S U = V \amalg W$$

into open and closed subschemes, such that $V \to U$ is finite and there exists a unique point v of V mapping to u, and such that v maps to t in T. As $V \to T$ is étale, we can lift generalizations, see Morphisms, Lemmas 03HV and 02GS. Hence there exists a specialization $v' \rightsquigarrow v$ such that v' maps to $t' \in T$. In particular we see that $v' \in X_U \subset T_U$. Denote $u' \in U$ the image of t'. Note that $v' \in \mathrm{Ass}_{X_U/U}(\mathcal{F})$ because $X_{u'}$ is a finite discrete set and $X_{u'} = \mathrm{Supp}(\mathcal{F}_{u'})$. As $\mathcal{F}$ is pure relative to S we see that v' must specialize to a point in X_u. Since v is the only point of V lying over u (and since no point of W can be a specialization of v') we see that $v \in X_u$. Hence $t \in X$. $\square$